

Implicit Learning Dynamics in Stackelberg Games Jin, C., Netrapalli, P., and Jordan, M. I. What is local optimality in nonconvex-nonconcave minimax optimization? arXiv preprint arXiv:1902.00618, 2019. Kelley, J. General Topology. Van Nostrand Reinhold Company, 1955. Kingma, D. P. and Ba, J. Adam: A method for stochastic optimization. In International Conference on Learning Representations, 2015. Knutson, A. and Tao, T. Honeycombs and sums of hermitian matrices. Notices of the American Mathematical Society, 2001. Kushner, H. J. and Yin, G. G. Stochastic approximation and recursive algorithms and applications, volume 35. Springer Science & Business Media, 2003. Lee, J. Introduction to smooth manifolds. Springer, 2012. Lee, J. D., Simchowitz, M., Jordan, M. I., and Recht, B. Gradient descent only converges to minimizers. In Conference on Learning Theory, pp. 1246–1257, 2016. Letcher, A., Foerster, J., Balduzzi, D., Rocktäschel, T., and Whiteson, S. Stable opponent shaping in differentiable games. In International Conference on Learning Representations, 2019. Maclaurin, D., Duvenaud, D., and Adams, R. Gradient-based hyperparameter optimization through reversible learning. In International Conference on Machine Learning, pp. 2113–2122, 2015. Madry, A., Makelov, A., Schmidt, L., Tsipras, D., and Vladu, A. Towards deep learning models resistant to adversarial attacks. In International Conference on Learning Representations, 2018. Mazumdar, E. and Ratliff, L. J. Local nash equilibria are isolated, strict local nash equilibria in ‘almost all’ zero-sum continuous games. In IEEE Conference on Decision and Control, 2019. Mazumdar, E., Jordan, M., and Sastry, S. S. On finding local nash equilibria (and only local nash equilibria) in zero-sum games. arxiv:1901.00838, 2019. Mazumdar, E., Ratliff, L. J., and Sastry, S. S. On gradient-based learning in continuous games. SIAM Journal on Mathematics of Data Science, 2(1):103–131, 2020. Mertikopoulos, P., Lecouat, B., Zenati, H., Foo, C.-S., Chandrasekhar, V., and Piliouras, G. Optimistic mirror descent in saddle-point problems: Going the extra (gradient) mile. In International Conference on Learning Representations, 2019. Mescheder, L., Nowozin, S., and Geiger, A. The numerics of gans. In Advances in Neural Information Processing Systems, pp. 1825–1835, 2017. Metz, L., Poole, B., Pfau, D., and Sohl-Dickstein, J. Unrolled generative adversarial networks. In International Conference on Learning Representations, 2017. Nagarajan, V. and Kolter, J. Z. Gradient descent gan optimization is locally stable. In Advances in Neural Information Processing Systems, pp. 5585–5595, 2017. Ortega, J. M. and Rheinboldt, W. C. Iterative Solutions to Nonlinear Equations in Several Variables. Academic Press, 1970. Panageas, I. and Piliouras, G. Gradient Descent Only Converges to Minimizers: Non-Isolated Critical Points and Invariant Regions. In Innovations in Theoretical Computer Science, 2017. Pemantle, R. Nonconvergence to unstable points in urn models and stochastic approximations. Annals Probability, 18(2):698–712, 04 1990. Radford, A., Metz, L., and Chintala, S. Unsupervised representation learning with deep convolutional generative adversarial networks. arXiv preprint arXiv:1511.06434, 2015. Ratliff, L. J., Burden, S. A., and Sastry, S. S. Genericity and structural stability of non-degenerate differential nash equilibria. In American Control Conference, pp. 3990–3995. IEEE, 2014. Ratliff, L. J., Burden, S. A., and Sastry, S. S. On the Characterization of Local Nash Equilibria in Continuous Games. IEEE Transactions on Automatic Control, 61(8):2301–2307, 2016. Sastry, S. S. Nonlinear Systems Theory. Springer, 1999. Schäfer, F. and Anandkumar, A. Competitive gradient descent. In Advances in Neural Information Processing Systems, pp. 7623–7633, 2019. Shub, M. Global Stability of Dynamical Systems. Springer-Verlag, 1978. Thoppe, G. and Borkar, V. A concentration bound for stochastic approximation via alekseev’s formula. Stochastic Systems, 9(1):1–26, 2019. Tretter, C. Spectral Theory of Block Operator Matrices and Applications. Imperial College Press, 2008. Von Stackelberg, H. Market structure and equilibrium. Springer Science & Business Media, 2010.